Conclusion

A predictive modeling project was developed using the Housing dataset to estimate house prices. Three machine learning algorithms were implemented and evaluated: Linear Regression, Decision Tree Regressor, and Random Forest Regressor.

The performance of each model was assessed using Mean Absolute Error (MAE) and R^2 Score.

Results:

- Linear Regression achieved an R^2 Score of 0.653 and performed best among all models.
- Random Forest achieved an R^2 Score of 0.612.
- Decision Tree achieved an R^2 Score of 0.477.

Based on the evaluation metrics, Linear Regression was selected as the most effective model for predicting house prices in this dataset.

This project provided practical experience in data preprocessing, supervised learning, model training, model evaluation, and performance comparison.